2313 44088 REGITA RASYA ZAFARANA 1C TNK

No
Date

1. a) Kuning = 4
Ungu = 7
Emas = 10^{-1}
Perak = 10%
Jadi, $47 \times 10^{-1} \pm 10\%$
$= 4,7 \times 10 \pm 10\%$
$4,7 \times 10\% = 4,7 \times \frac{10}{100} = 0,47\ \Omega$
Toleransi max = 4,7 + 0,47 = 5,17 Ω
min = 4,7 - 0,47 = 4,23 Ω

b) Orange = 3
putih = 9
kuning = 10^4
Emas = 5%
Jadi, $39 \times 10^4 \pm 5\%$
$= 390k\Omega \pm 5\%$
$390.000 \times 5\% = 19,5k\Omega$
Toleransi max = 390K + 19,5K = 409,5 KΩ
Toleransi min = 390K - 19,5K = 370,5 KΩ

c) coklat = 1
hitam = 0
merah = 10^2
Emas = 5%
Jadi, $10 \times 10^2 \pm 5\%$
$= 1000\ \Omega \pm 5\%$
$1000 \times 5\% = 50\ \Omega$
Toleransi max = 1000 + 50 = 1050 Ω
Toleransi min = 1000 - 50 = 950 Ω

231344088 REGITA RASYA Z 1C TNK

No
Date

d) Coklat = 1
hijau = 5
hijau = 10^5
Tidak berwarna = 20%
Jadi, $15 \times 10^5 \Omega \pm 20\%$
$= 1500 k\Omega \pm 20\%$
$1.500.000 \times 20\% = 300 k\Omega$
Toleransi max = $1500 k + 300 k = 1800 k\Omega$
Toleransi min = $1500 k - 300 k = 1200 k\Omega$

e) hijau = 5
biru = 6
hitam = 0
merah = 10^2
coklat = 1%
Jadi, $560 \times 10^2 \Omega \pm 1\% = 56 k\Omega \pm 1\%$
$56.000 \times 1\% = 56000 \times \frac{1}{100} = 560 \Omega$
Toleransi max = $56000 + 560 = 56.560 \Omega$
Toleransi min = $56000 - 560 = 55.440 \Omega$

f) Coklat = 1
Abu-abu = 8
hitam = 0
hitam = 10^0
coklat = 1%
Jadi, $180 \Omega \pm 1\%$
$180 \times 1\% = 1,8 \Omega$
Toleransi max = $180 + 1,8 = 181,8 \Omega$
Toleransi min = $180 - 1,8 = 178,2 \Omega$

No
Date

2) a) $R = 100\ \Omega$

$I = 1000\ mA = 0,1\ A$

V? $V = I \cdot R$

$= 0,1 \times 100$

$= 10\ V$

P? $P = I^2 \times R$

$= (0,1)^2 \times 100$

$= 1\ W$

b) $R = 1\ k\Omega$

$V = 5\ V$

I? $I = \frac{V}{R}$

$= \frac{5}{1000}$

$= 0,005\ A$

P? $P = I^2 R$

$= (0,005)^2 \times 1000$

$= 0,025\ W$

c) $I = 1\ A$

$V = 5V$

$R = \frac{V}{I}$

$= \frac{5}{1}$

$= 5$

P? $P = 1^2 \times R$

$= 1^2 \times 5$

$= 5\ W$

231344088 REGITA RASYA Z 1C TNIC

No
Date

3.) a) Dik : V = 5
I = 10 mA = 0,01 A
Vled = 2V
Dit : Rs ?
Jawab : $R = \frac{(V_s - V_L)}{I}$
$= \frac{(5-2)}{0,01}$
$= \frac{3}{0,01}$
= 300 Ω → 330 Ω
Dikarenakan pada E12 yang paling mendekati antara 2,7 dan 3,9 adalah 3,3

b) Dik : V = 12 V
I = 1 mA = 0,001 A
$V_{R1} = \pm 7V$
$V_{R2} = \pm 5V$
Dit : $R_1 = ?$
$R_2 = ?$
Jawab : $R_1 = \frac{V_s - V_1}{I}$
$= \frac{12-7}{0,001}$
$= \frac{5}{0,001}$
= 5000 Ω
= 5 kΩ → 5,6 kΩ
$R_2 = \frac{V_s - V_2}{I}$
$= \frac{12-5}{0,001}$
= 7000 Ω
= 7 kΩ → 8,2 kΩ

4) a) $R_T = R_1 + R_2$
= 10Ω + 10Ω
= 20Ω
P_{total} = 1W + 1W
= 2W

b) $R_T = \frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2}$
$= \frac{1}{10} + \frac{1}{10}$
$= \frac{2}{10}$
$\frac{R_T}{1} = \frac{10}{2} = 5\Omega$
P_{total} = 1W + 1W
= 2W

231344088 REGITA RASYA Z 1C-TNK

No
Date

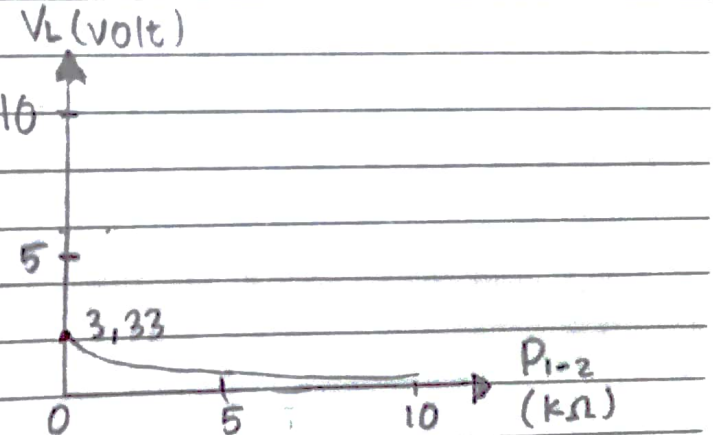
c) $R_{S1} = R_1 + R_2$
$= 10\Omega + 10\Omega$
$= 20\Omega$

$P_{total} = P_{RS1} + P_{RS2}$
$= 2W + 2W$
$= 4W$

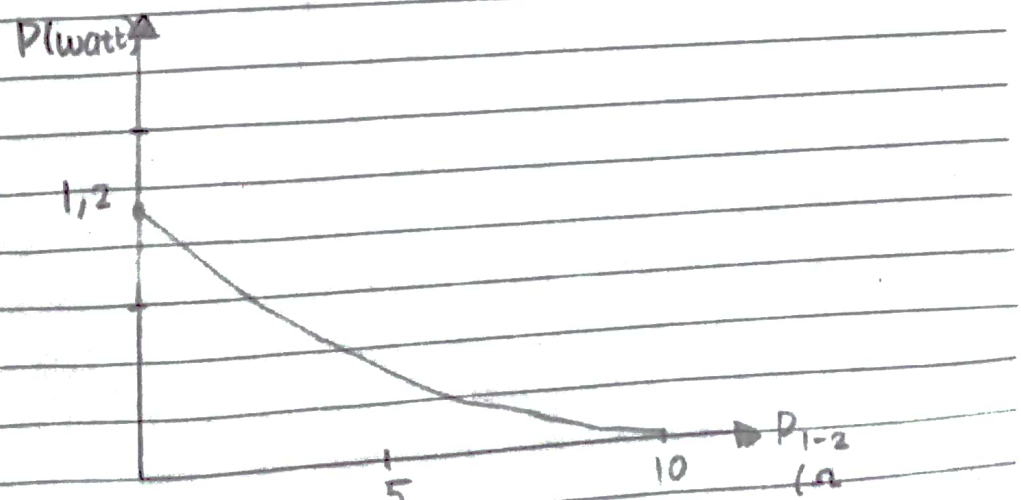
$R_{S1} // R_{S2} = \frac{1}{R_{S1} // R_{S2}}$
$= \frac{1}{20} + \frac{1}{20}$
$= \frac{2}{20}$
$R_{S1} // R_{S2} = \frac{20}{2} = 10\Omega$ //

5) a) Jika $R_L = \infty$ maka $V_L = 0$, karena rangkaian akan bersifat open circuit, maka grafik untuk kondisi ini akan sejajar pada sumbu P_{1-2} (kΩ) dikarenakan $V_L = 0$ dan $R_L = \infty$

b) Dik : $V_3 = 10V$
$P_1 = 10k$
$RL = 5k$
Jawab: $V_L = \frac{RL}{P_1 + RL} \times V_s$
$V_L = \frac{5k}{5k+10k} \times 10$
$= 3,33\ V$



6) $V = 12\ V$
$R_1 = 5\Omega$
$P = 10\Omega$
$R_1 = \frac{V}{P}$
$= \frac{12}{10}$
$= 1,2\ W$



231344088 REGITA RASYA Z 1C- TNK

No
Date

7) Dik : $V_o = 5V$

LDR tahanan gelap = $2M\Omega$

LDR tahanan terang = 300Ω

RB => R Bottom

RT => R Top

Dit : besar nilai R_1 ?

Jawab : $V_{out} = \frac{R_B}{R_B + R_T} \times V_{in}$

$5V = \frac{R_B}{R_B + 300\Omega} \times 12V$

$5V(R_B + 300) = R_B \times 12V$

$R_B + 300 = \frac{R_B \times 12\cancel{V}}{5\cancel{V}}$

$R_B + 300 = R_B \times \frac{12}{5}$

$\frac{300}{R_B} = \frac{12}{5}$

$300(5) = 12\,R_B$

$\frac{1500}{12} = R_B$

$125\,\Omega = R_B$